

Enterprise network planning and design: methodology and application

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Abstract

The paper describes a business-driven enterprise network planning and design methodology that has an iterative structure to periodically assess, update, and extend the network architecture to support new business applications and requirements, account for changes in business direction, and analyze the implications of technology advances on business processes and functions. The paper focuses on the network infrastructure design and analysis task of the proposed methodology and discusses the network design and analysis of a multi-protocol application environment characterized by a client/server architecture, integration of legacy and distributed database structures, frame relay wide area network, and token ring local area networks. Detailed workload characterization and performance modeling of the application environment is presented, and network design issues are addressed. The model is used to study network performance under various load conditions, identify bottlenecks and compare design alternatives. © 1999 Elsevier Science B.V. All rights reserved.

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1. Introduction

The objectives of enterprise network planning are to develop a network architecture that supports the enterprise's business direction and requirements, and to identify the strategies that need to be followed and the actions that need to be taken to close the gap between the current and the desired state of the network architecture. To address these objectives, we propose a structured network planning process, shown in Fig. 1, that extends traditional network planning concepts to include business, industry, market, and regulatory factors along with service and technology factors. The process starts with an analysis of the business requirements and revenue opportunities, instead of cost-reduction technology choices as is typically the case in technology driven approaches. The planning methodology is iterative, and one of its main components is the *network architecture management process* which is used to periodically assess, update, and extend the network architecture to support new business applications and requirements, account for changes in business direction, and analyze the implications of technology advances on business processes and functions. The main tasks of the methodology are:

1.1. Strategic business modeling

This task analyzes business requirements and revenue

opportunities, and identifies the business applications/services that need to be supported by the network infrastructure [1].

1.2. Industry and technology trends analysis

This task identifies and analyzes the implications of technology advances and timelines in terms of time of introduction, maturity, acceptability, and standards, on business processes, functions, and applications.

1.3. Network strengths, weaknesses, opportunities and threats analysis

This task assesses the corporation's usage of network technology, and develops an inventory of the current network infrastructure. The task identifies the strengths, and weaknesses of the current network infrastructure, the opportunities to apply technology to enhance network strategy, and the barriers to successfully utilize network solutions to support business directions.

1.4. Network architecture planning

This task develops network architectures that take advantage of technology capabilities to efficiently support business application requirements. This phase of network

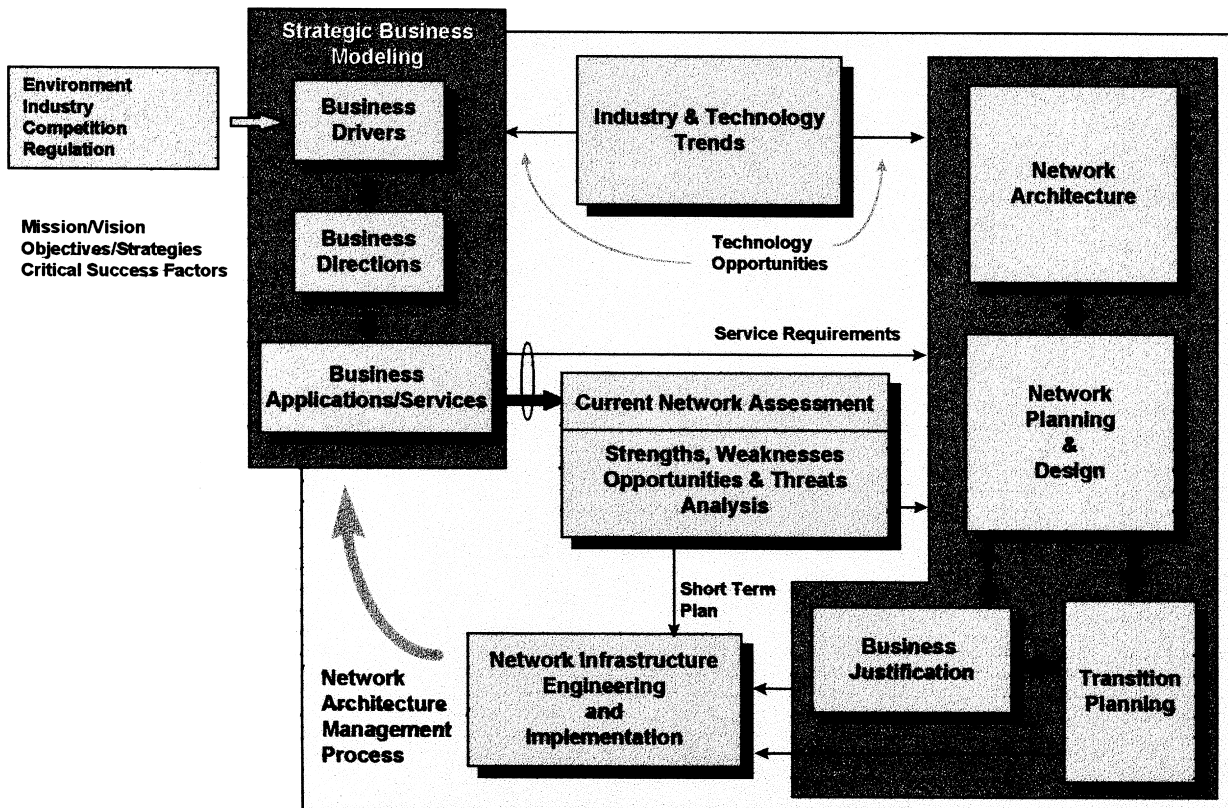


Fig. 1. Strategic network planning process.

architecture planning is de-coupled from the physical network implementation. The emphasis is on developing functional architecture models that specify the key functions that need to be performed and their interactions.

1.5. Network planning and design

This task determines short, medium, and long term

network plans, for technology deployment according to the defined network architecture, and uses optimization techniques to minimize the total network infrastructure cost while meeting all constraints and design requirements.

1.6. Business justification and transition planning

This task identifies the strategies that will be followed and

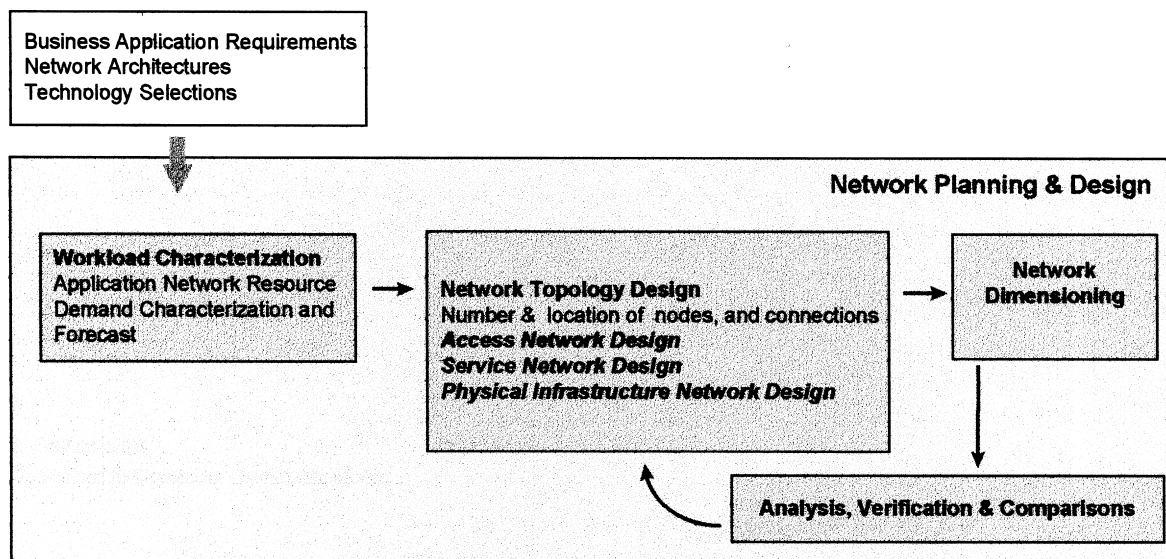


Fig. 2. Network planning and design process.

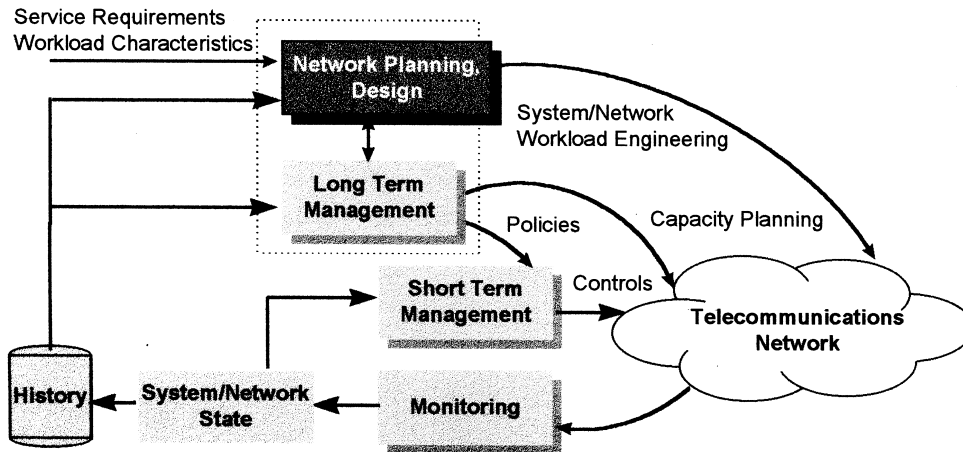


Fig. 3. Performance management process.

the actions that will be taken to close the gap between the current and the desired state of the corporate network infrastructure. The task uses a variety of engineering economic methods and tools [2–4] to evaluate alternatives and provide business justification for network technology investment recommendations.

1.7. Network infrastructure engineering and implementation

This task addresses detailed engineering and network infrastructure deployment and implementation issues.

The main activities associated with the network planning and design task, shown in Fig. 2, are: network requirements specification, workload characterization, network topology design, network dimensioning, analysis and comparison of

design options, and network design verification. The purpose of the workload characterization activity is to identify the traffic characteristics and network resource requirements of the current and planned business applications. The network topology design activity, addresses network optimization issues related to the number and location of network nodes and the connections between them, taking into account the constraints imposed by the existing network infrastructure. The network dimensioning task uses performance and reliability modeling and optimization methods to determine the optimum network configuration based on a given network architecture and technology selection, and traffic load characteristics. Finally, the various configurations are analyzed and evaluated. This activity is iterative, and involves sensitivity analysis that allows for the modification of design parameters and the selection of new scenarios.

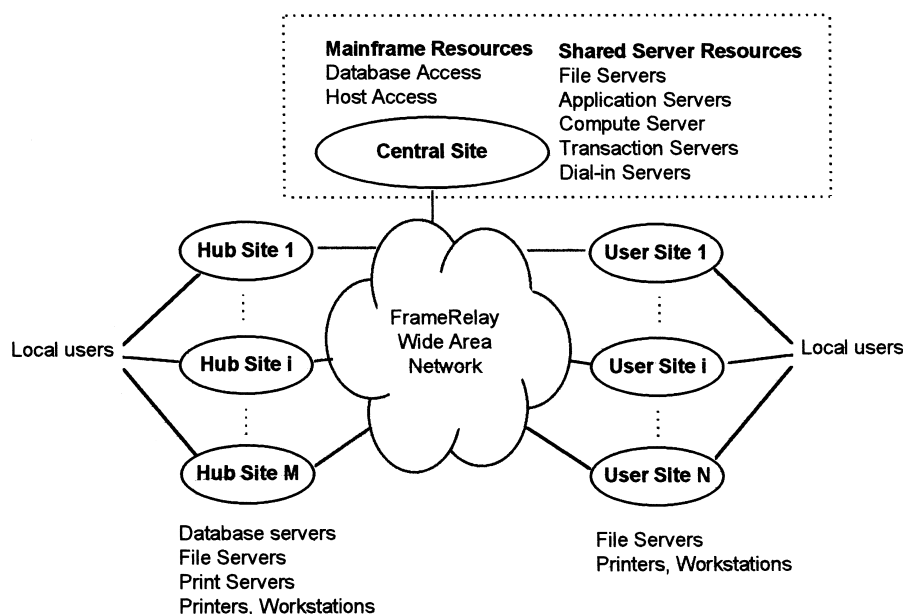


Fig. 4. High level view of the application environment.

The network planning and design task, is an integral part of the network performance management process, as shown in Fig. 3, where performance management is described via a control model whose basic feature is a feedback process that consists of multiple feedback loops that operate at various time intervals and control various aspects of the network behavior. The main tasks associated with the performance management process are the:

- *monitoring activity*, enables the collection of measurements that provide a central view of the current state and over time the network history;
- *short term activities*, that take place in time scales of minutes or hours, tend to be reactive in nature and are aimed at detecting problems, identifying their cause and fixing them;
- *long term activities*, take place in time scales of days, weeks or longer, and are aimed at ensuring that the network architecture can efficiently support the application workload requirements.

The performance management process is supported by tools that incorporate performance modeling, simulation, optimization, and artificial intelligence techniques, among others.

The paper uses the proposed methodology to plan and design a private network infrastructure that enables a corporation to automate key business functions, and improve its efficiency and responsiveness to changing business needs. The focus of the study is on the network infrastructure design and analysis task of the methodology and specifically on the network dimensioning activity. The organization of the paper follows. Section 2 describes the application environment under consideration and the network requirements. Section 3 presents the workload characterization. System performance modeling and evaluation is presented in Section 4. Section 5 addresses various network design issues. Finally, Section 6 presents some concluding remarks.

2. Requirements specification

The application environment under consideration is shown in Fig. 4. The environment provides users with transparent access to shared server and database resources that support mission critical business applications. The shared resources are located at a number M of *hub sites* and a *central site*.

The hub sites represent regional centers, that are strategically placed at various geographical locations across the entire coverage region. Each hub site i ; $i = 1, \dots, M$, provides database, file and print services to L_i local users, and to K_i user sites each with Rj_i ; $j_i = 1, \dots, K_i$, $i = 1, \dots, M$, users (for a user site j_i the subscript i is used to identify the hub site to which the user site is assigned). User sites represent local offices strategically located across the coverage region to

optimize customer access. A wide area network is used to interconnect the $N = \sum_i K_i$ user sites, the M hub sites and the central site. Each user site j_i ; $j_i = 1, \dots, K_i$, supports printer and workstation resources and optionally provides file and print services to Rj_i local users. Database services and optionally file and print services, are provided by the hub site i , $i = 1, \dots, M$ to which the user site is assigned. The central site provides services (database, file, application, compute, transaction, dial-in) to all hub and user sites, through a combination of shared server and mainframe resources.

The study focuses on the network design, analysis, and engineering of the application environment under consideration. The network design is based on a simulation model that is used to study network performance under various load conditions, identify potential bottlenecks and compare design alternatives. The main network design requirements for the application environment under consideration are as follows:

- A frame relay wide area network will be used to interconnect the $N = \sum_i K_i$ user sites, the M hub sites, and the central site.
- Each hub site i supports a number K_i of user sites. Users in the user sites access application environment resources through their assigned hub site.
- There is no direct communication between user sites.
- Hub sites should be able to communicate with each other, but the hub-to-hub site traffic is very low, and does not cost-justify the use of direct frame relay permanent virtual circuits (PVCs) between hub sites. Communication between hub sites should be supported by indirect paths through the central site.
- The application environment uses a distributed heterogeneous database structure. Database services are provided by the central site and each hub site. The database manager at each hub site provides transparent access to the various databases for all users at the hub site and the supported user sites. Access to the mainframe legacy database is provided through a database gateway located at the central site. The legacy database uses the SNA communication protocols. All other databases use the TCP/IP protocols.
- All local area networks are token ring based. The network operating system used to provide file and print services, is Novell's NetWare [10].
- Mainframe access is provided through a 3270 emulation software running on each user workstation and a NetWare for SAA gateway [11]. Communication between the user workstations and the gateway is based on SPX/IPX. The gateway reformulates each SPX/IPX packet as an SNA packet which is then bridged over the appropriate network.

The location of the various sites (user, hub, and central) depends on business factors, and for the purposes of this study, is assumed to be known. The number of users per

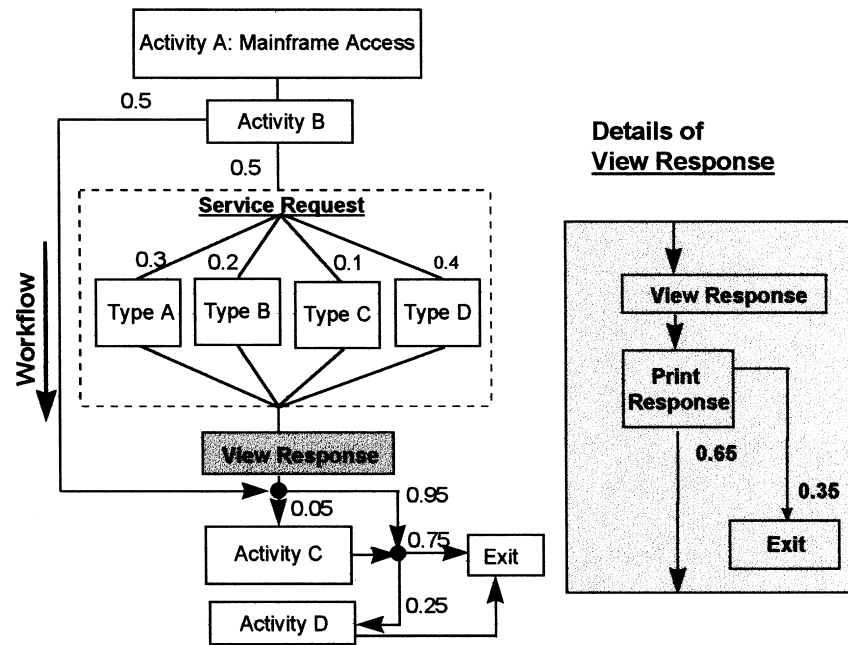


Fig. 5. Functional level application workload characterization.

site and the number of sites are considered to be design variables. The technology choices mentioned in the requirements, namely communication protocols (TCP/IP, SPX/IPX, SNA, Frame Relay), network operating system, and various gateways, depend on a number of business and embedded infrastructure considerations. The justification of these technology choices, is outside the scope of the study.

3. Workload characterization

The application environment is used to automate and improve the efficiency, flexibility and performance of a business function supported by the process shown in Fig. 5. The business process is described through a number of activities (generically referred to as activity A, B, C and D, in the figure), that use network and computing resources located at the user, hub and central sites, to perform the business function. In the following we use the term *job* to collectively refer to the application workload associated with all the activities executed to support the business

process shown in Fig. 5. The jobs are interactive and are executed by users located at the various user or hub sites. The workload intensity is described in terms of the number of simultaneously executing jobs in the system.

As shown in Fig. 5, each job execution involves activity A (mainframe access) and activity B. After completing the execution of activity B, the user may request a shared service from the central site shared server resources with a 50% probability. The service requested by the user can be one of four types (the probability distribution is shown in the figure). After completion of the service request, the user views and prints the results, and may complete job execution with a 35% probability, or continue with a 65% probability, to execute activities C and D according to the conditions shown in the figure.

A detailed workload characterization was performed for each activity associated with the job execution. Due to space limitations, in the following we discuss the workload characteristics of the first activity only. The main functions related to activity A, *mainframe access*, and the associated network traffic, are summarized in Table 1. In the table we show for each function, the traffic (in number of bytes)

Table 1
Activity A: mainframe access

Function	Request/query size (bytes)		Response size (bytes)	
	User–hub traffic	Hub–mainframe traffic	Hub–user traffic	Mainframe–hub traffic ^a
Load application	10	—	200,000	—
Transactions (15/user)	16	16	350–500	350–500
On-line report	—	—	154,960	154,960
Transfer to user site	—	—	154,960	—
Print report	154,960	—	154,960	—

^a All mainframe database related traffic is routed through the database gateway located at the central site.

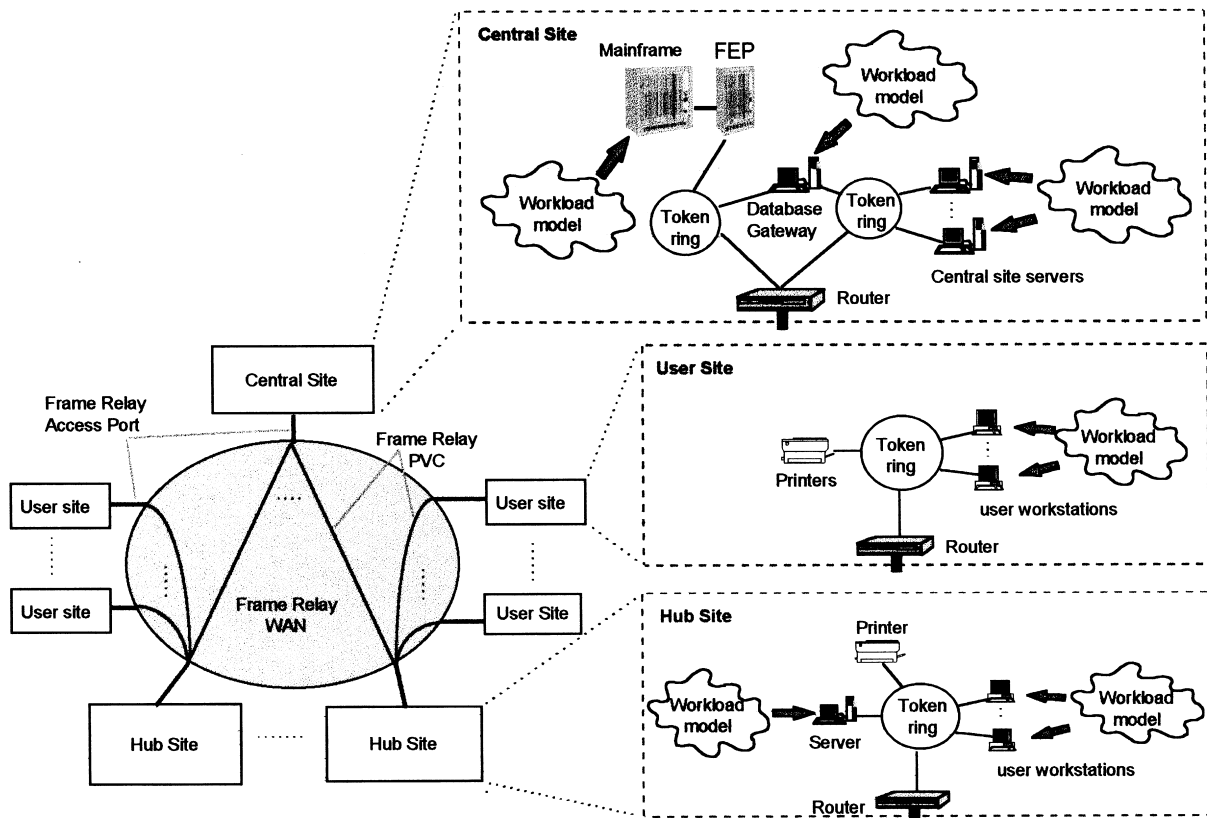


Fig. 6. High level structure of the application environment model.

between the user and hub sites, and between hub sites and the mainframe resources in the central site. The first function associated with the activity, is to load the mainframe access application. The location of the application binaries is a design variable. Two alternatives are considered: the binaries are stored on physical storage devices residing either in the hub site or in the user site. After loading the application, the user performs various functions that involve 15 mainframe transactions. For each transaction the request size is 16 bytes and the average response size 350–500 bytes. Once a day every user will print a report, that consists of an average of 104 screens with 1490 bytes/screen, which implies that the average report size is 154,960 bytes. To print the report, the entire report is transferred from the mainframe to the user site workstation. If the print server is located in the hub site, then the report must be sent from the user site to the print queue on the hub file server, and then to the user site printer.

A similar analysis is performed for each activity shown in Fig. 5. The results of the workload characterization are used to develop a detailed workload model that captures the application workload characteristics.

4. Performance modeling and analysis

To study the performance of the application environment

under various load conditions, to identify bottlenecks and compare design alternatives, we developed a simulation model of the application workload and the network infrastructure. The model consists of two major components, the resource model and the workload model. The resource model represents the various hardware resources and the mechanisms/protocols used for resource allocation and control. The following resources are explicitly modeled:

- Frame Relay Wide Area Network. TCP/IP encapsulation is used to transport SNA traffic over the frame relay wide area network.
- Token Ring Local Area Networks at the user sites, hub sites and the central site (there are two token ring LANs in the central site, one for the shared server resources and one for the mainframe host).
- Routers at the user sites, hub sites and the central site.
- Communication protocols (TCP/IP, SPX/IPX, and SNA). The network operating system used to provide file and print services, is Novell's Netware, which uses the SPX/IPX protocols. Communication between the various hub site databases and the central site shared server resources is based on TCP/IP protocols. SNA is used for mainframe based transactions. Access to the mainframe database is accomplished through a database gateway located at the central site. TCP/IP protocols are used for communication between hub site databases and

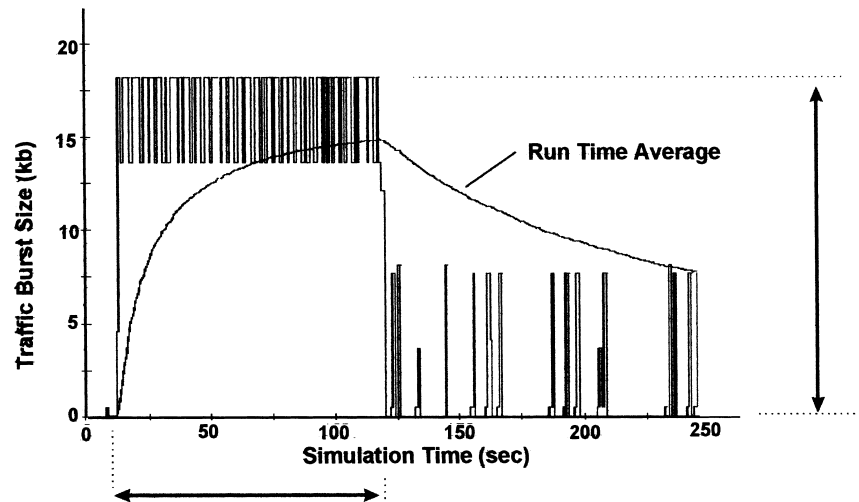


Fig. 7. Burst size (in kb) for the traffic across the frame relay PVC (hub site to the user site direction) generated by the operations associated with activity B (refer to Fig. 4). The file server is located at the hub site. The workload is generated by one user at the user site. The NCP data unit size is equal to 512 bytes and the window size is equal to 1.

the database gateway. Communication between the database gateway and the mainframe database is based on SNA.

- Routing protocols. The application environment uses OSPF [5–7] to route IP, and RIP [7–9] to route IPX.

The servers, mainframe, database gateway, and user workstations are modeled as delay centers, to simulate the effect that processing delay on these resources has on the intensity of the workload that is applied to the network.

The workload model captures the characteristics of the application workload that is applied to the network. The workload model is based on detailed traffic measurements collected during application benchmarks (the workload

characteristics are described in Section 2). To model the user think time (for example the time spent reviewing the information on a application screen), we use delay centers. All think times are assumed to be exponentially distributed.

A high level diagram of the models used to represent the frame relay wide area network, the central site, the hub sites, and the user sites, are shown in Fig. 6. In the following we discuss simulation model results and frame relay wide area network design issues.

4.1. User site performance modeling results

In this subsection we analyze the effect of the workload intensity (expressed in terms of number of simultaneously

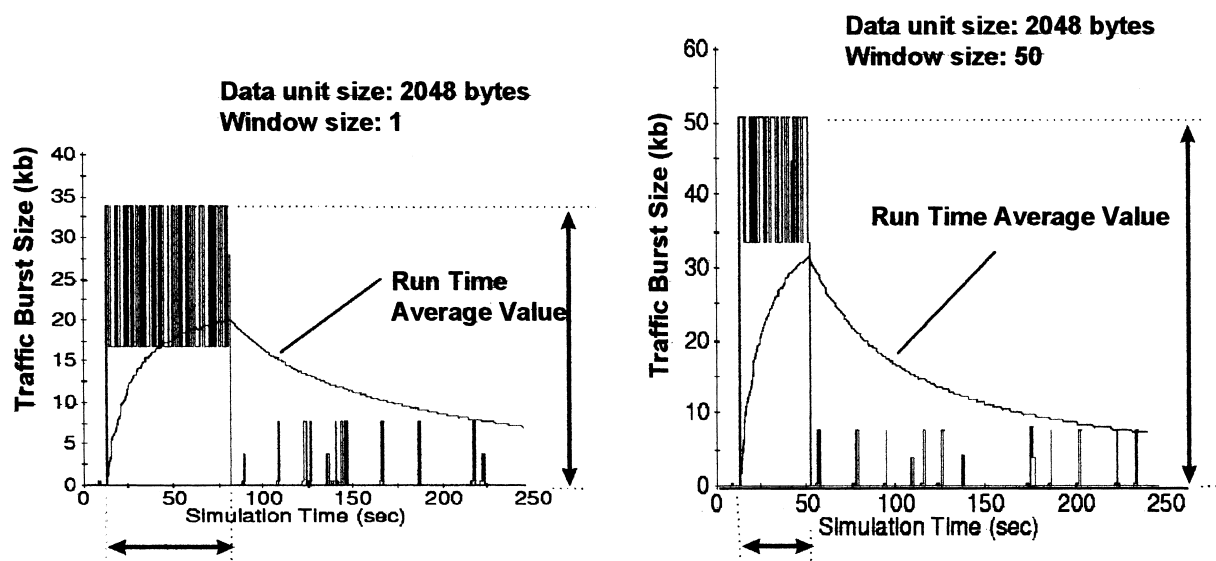


Fig. 8. Burst size (in kb) for the traffic across the frame relay PVC (in the hub site to the user site direction) generated by the operations associated with activity B (refer to Fig. 4). The file server is located at the hub site. The workload is generated by one user at the user site. The NCP data unit size is equal to 2048 bytes and the window size is equal to 1 and 50.

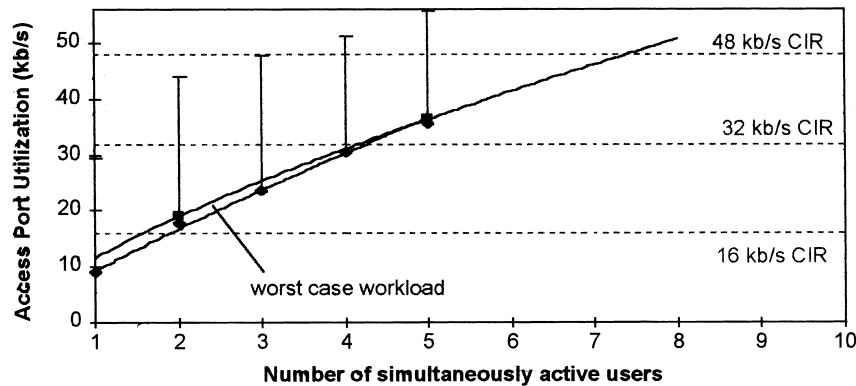


Fig. 9. User site frame relay access port utilization (in kb/s) due to the traffic from the hub site to the user site, as a function of the number of simultaneously active users. The file server is located at the hub site. Starting from the bottom, the first curve shows the average access port utilization for typical application workload; the second curve shows the average access port utilization for a worst case application workload. The bars show the standard deviation of the access port utilization.

active users at the user site), network protocol configuration settings and file server location, on the wide area network bandwidth resource requirements and system performance. Initially we consider a configuration where all shared services¹ (file, print, and database) are provided by the hub site server. We use the term *remote applications* when referring to this configuration.

The main protocol configuration parameters that affect the performance of file and print services supported by the application environment, are the application protocol buffer (or data unit) size and the window size of the flow control protocol². To illustrate the effect of these two parameters on the traffic flow across the wide area network, we show in Fig. 7, the traffic across the hub-to-user site frame relay PVC, generated by the functions associated with a typical user activity (which in this case is assumed to be activity B from Fig. 4). The data unit is set to 512 bytes and the window flow control size to 1.

From the figure we note the traffic burst on the left side of the graph, associated with the loading of the application executables, and the subsequent traffic spikes due to the hub and user site interactions. The small value of the data unit size and window flow control size, result in a traffic burst characterized by large burst duration (~ 120 s) and moderate burst size (~ 17 kb). Increasing the data unit size to 2048 bytes, while keeping the window size equal to one, results in a reduction of the burst duration and an increase of the burst size, as shown in Fig. 8. Keeping the data unit size at 2048 bytes and increasing the window size to 50, results in a further reduction of the burst duration to approximately

50 s, as shown in Fig. 8, but during this period the 56 kb/s port utilization is very high. In the following we consider a configuration with a data unit size equal to 2048 bytes, and a window size equal to one.

In Fig. 9, we show performance results for the remote applications configuration. Specifically, we show the user site frame relay access port utilization as a function of the number of simultaneously active users in the user site. The typical activities performed by each user are shown in the workflow chart in Fig. 5. To study the impact of network resource intensive functions on overall system performance, in Fig. 9 we also show performance results for a worst case application workload that is characterized by an increased frequency of execution of certain network resource intense functions (such as the on-line report view/print function mentioned in Section 2). The access port utilization results in Fig. 9, indicate that the remote applications configuration generates heavy and bursty traffic between the hub site and the user site. This is also shown in Fig. 10 by the time series plots of the frame relay access port utilization for a user site with two and five simultaneously active users.

To study the effect of the file server location on system performance, we consider a configuration where the application executables are located at the user site, either on a file server or on the workstation's hard drive. We use the term *local applications* when referring to this configuration. Database service and print queuing are provided by the hub site server.

In Figs. 11 and 12, we plot the user site frame relay access port utilization and traffic burst size, as a function of the number of active users in the user site. The performance results shown in the figure, are associated with the traffic that flows across the frame relay PVC, in the hub-to-user site direction. The traffic flow between the user site and the hub site is asymmetric, with the traffic in the hub-to-user site direction substantially heavier than the traffic in the user-to-hub site direction. Due to space limitations, we only show results for the hub-to-user site direction. Comparing the

¹ These do not include the shared services provided by central site resources.

² The network operating system used by the application environment to provide file and print services, is Novell's Netware, which uses the Network Control Protocol (NCP) over the SPX/IPX protocol stack [10]. The two NCP configuration parameters that affect system performance and traffic burstiness, the NCP data unit size and the window size of the flow control protocol.

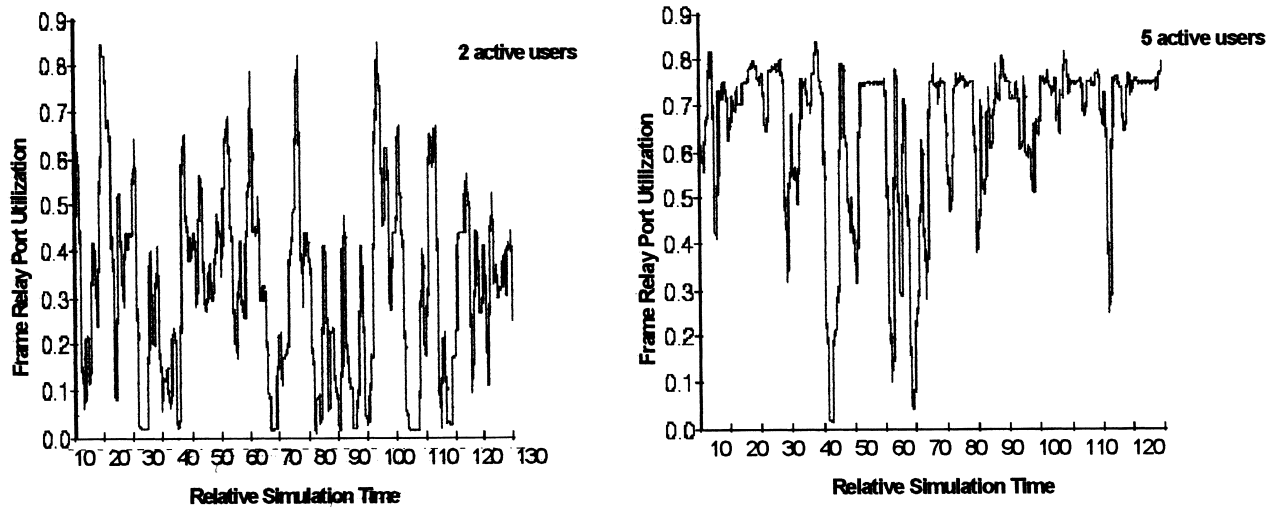


Fig. 10. User site frame relay access port utilization (expressed as the fraction of the 56 kb/s port capacity used) due to the traffic from the hub site to the user site, as a function of simulation time. The file server is located at the hub site. The application workload is generated by two (upper figure) and five (lower figure) active users at the user site.

local and remote application configuration results, we note the reduction in frame relay access port utilization and traffic burstiness achieved with the local applications configuration, due to the reduced number of user-to-hub site interactions and file transfers across the wide area network. Finally, in Fig. 13, we show the impact of the file server location on the frame relay data unit delay across the PVC connecting the hub site and the user site.

4.2. Hub site performance modeling results

To determine the required hub site frame relay access port rate, we plot in Fig. 14 the hub site frame relay access port utilization as a function of the number of simultaneously active users in the user sites supported by the hub site. Results are shown for both *remote* and *local applications* configurations. Specifically, three performance curves are shown for each configuration, that represent the following

(starting from the lower curve and moving upwards within each set of curves):

- Average port utilization (for a 256 kb/s port configuration) assuming no active local users at the hub site. The average port utilization for a 128 kb/s port configuration, is also shown.
- Average port utilization (for a 256 kb/s port configuration) assuming 40 local users at the hub site.
- Standard deviation of the port utilization.

From the performance curves shown in the figure we note that the *remote applications* configuration has a substantially higher frame relay access port utilization compared to the *local applications* configuration, limiting scalability and overall system performance.

The performance results shown in Fig. 14 are used to engineer the hub site frame relay access port rate, and the CIRs for the PVCs connecting the hub site to the supported user sites. As evident from the figure, the port rate and the PVC CIRs depend on a number of parameters: the number

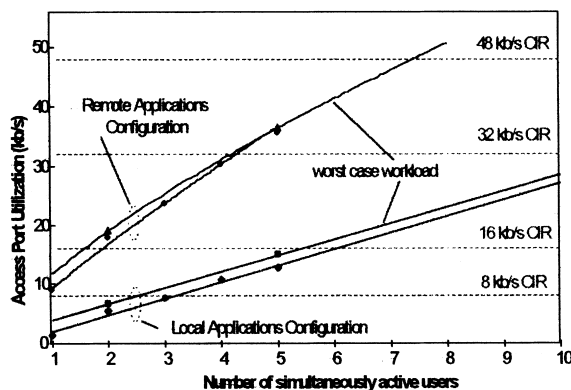


Fig. 11. User site frame relay access port utilization (in kb/s) due to the traffic from the hub site to the user site, as a function of the number of simultaneously active users.

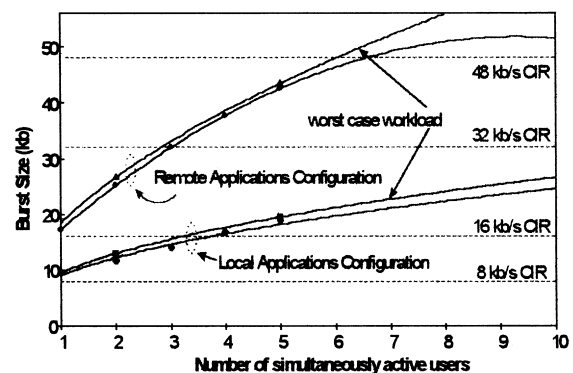


Fig. 12. Traffic burst size (in kb) for the hub-to-user site traffic, as a function of the number of simultaneously active users in the user site.

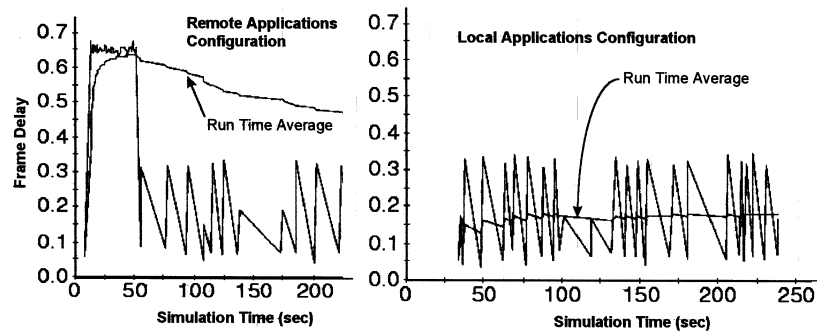


Fig. 13. Time series plot of the frame relay data unit delay across the PVC connecting the hub site and the user site, for remote and local applications configurations.

of supported user sites per hub site, the number of users at each supported user site, the user site configuration (local versus remote applications) and the number of local users at the hub site.

4.3. Central site performance modeling results

To determine the required central site frame relay access port rate and the CIR of the PVC between the central site and each hub site, in Fig. 15 we plot the central site frame relay access port utilization as a function of the number of users supported by a hub site. Three performance curves are shown in the figure. The first curve starting from the bottom and moving upwards, is associated with an application workload that consists of typical application environment operations. For this workload the central site access port utilization is low. The situation is different for the worst case application workload, which is characterized by functions that require substantial information exchange between

the central and hub site resources. This is illustrated by the middle and upper curves, that show the average and standard deviation, respectively, of the central site frame relay access port utilization, for a worst case application workload.

5. Network design issues

In this section we discuss various network design and engineering issues related to the application environment under consideration. The location of the application executables, and of the NetWare for SAA gateway, is a design variable. The following alternatives are considered:

- *Application executables*: two options are considered for the location of the application executables: (a) locate at the hub file server and provide access to the user sites via the frame relay wide area network; (b) locate at the user

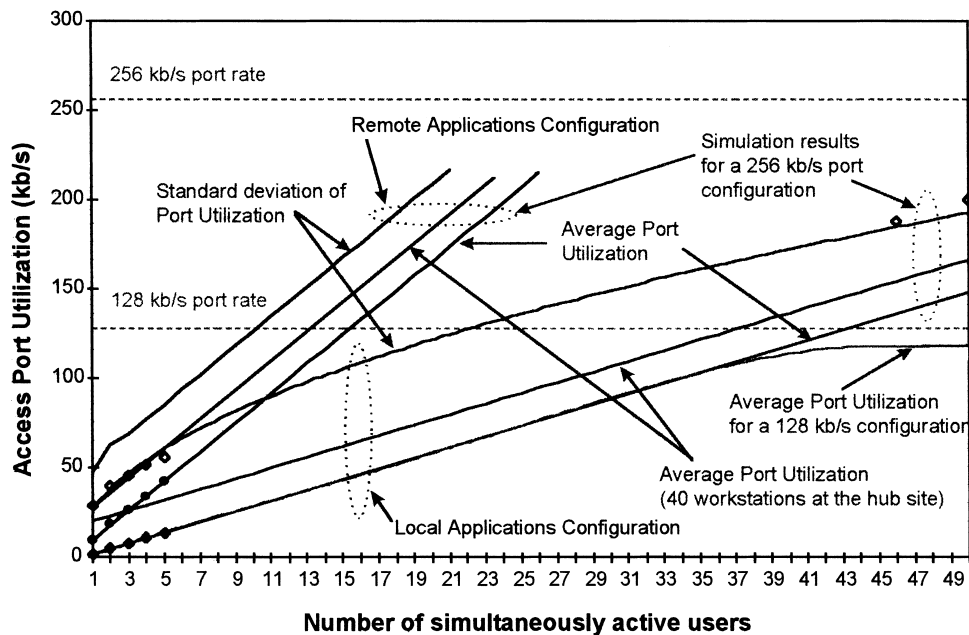


Fig. 14. Hub site frame relay access port utilization (in kb/s) due to the hub-to-user sites traffic flow, as a function of the number of simultaneously active users at the user sites supported by the hub site. Performance results are shown both for *remote* and *local applications* configurations.

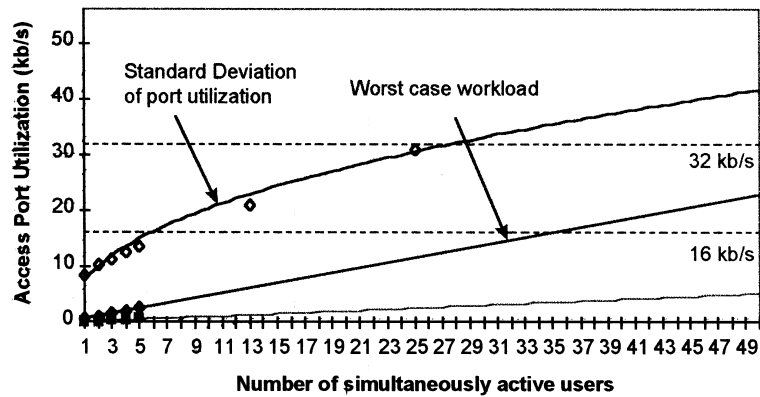


Fig. 15. Hub site frame relay access port utilization (in kb/s), as a function of the number of simultaneously active users. The curve in the lower part of the graph shows average access port utilization for typical application environment operations. The other two curves show the average and the standard deviation of the access port utilization for worst case application workload.

site, either on a user site file server, or on each workstation's hard drive. The first option has a cost advantage as shown in Fig. 16, since it requires a substantially smaller number of hardware resources, but based on the analysis presented in the previous section, it has a significant disadvantage in terms of overall network performance and scalability. The second option can be implemented by using either local file servers or by storing files on workstation's hard drives. The file server approach, simplifies software distribution, but it has a higher cost. A combination of the two approaches can be used depending on the number of workstations per user site.

- *NetWare for SAA gateway*: two options are considered for the location of the NetWare for SAA gateway [11]: (a) the gateway runs on the hub site file server; (b) the gateway runs on a dedicated server at the central site. The first option requires transport of SNA traffic across the frame relay wide area network, which increases the complexity of the router configuration and the router processing requirements. The second option simplifies the overall network design, since only routed protocols (IPX and IP) need to be supported across the frame relay wide area network. Furthermore, it enables centralized management and administration of the NetWare for SAA

configuration. A disadvantage with this approach is the increase in the number of sessions that will be affected in the case of gateway failure.

To transport SNA traffic across the wide area network, we use *Data Link Switching (DLSw)* which is a standard protocol for transporting SNA traffic over an IP network [12]. The standard specifies the Switch-to-Switch Protocol (SSP), which is used between routers to establish DLSw connections, locate resources, forward data, and handle flow control and error recovery. Two basic characteristics of DLSw that are of particular interest to our discussion are the TCP/IP transport and the local termination of DLC connections. DLSw specifies TCP/IP as the standard mechanism for transport of SNA traffic across the wide area network. It also describes how to terminate DLC connections locally and map the DLC connections to the TCP connection. The major benefit of the TCP/IP encapsulation is a robust set of capabilities to ensure reliable transport. TCP/IP encapsulation imposes a high cost in terms of router processing overhead because it involves a number of tasks such as packet reordering, running timers for retransmission, and sending acknowledgments. The local termination of DLC connections at each DLSw router eliminates all potential data link session time-out problems, and prevents

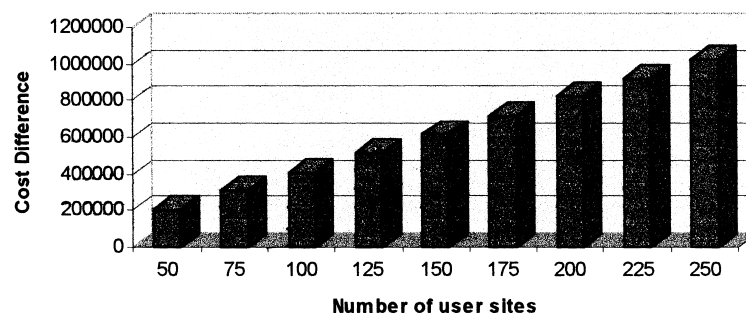


Fig. 16. Cost difference (in US dollars) between the local and remote applications configuration (cost of local configuration minus cost of remote configuration) as a function of the number of user sites. An average of five users per user site is assumed.

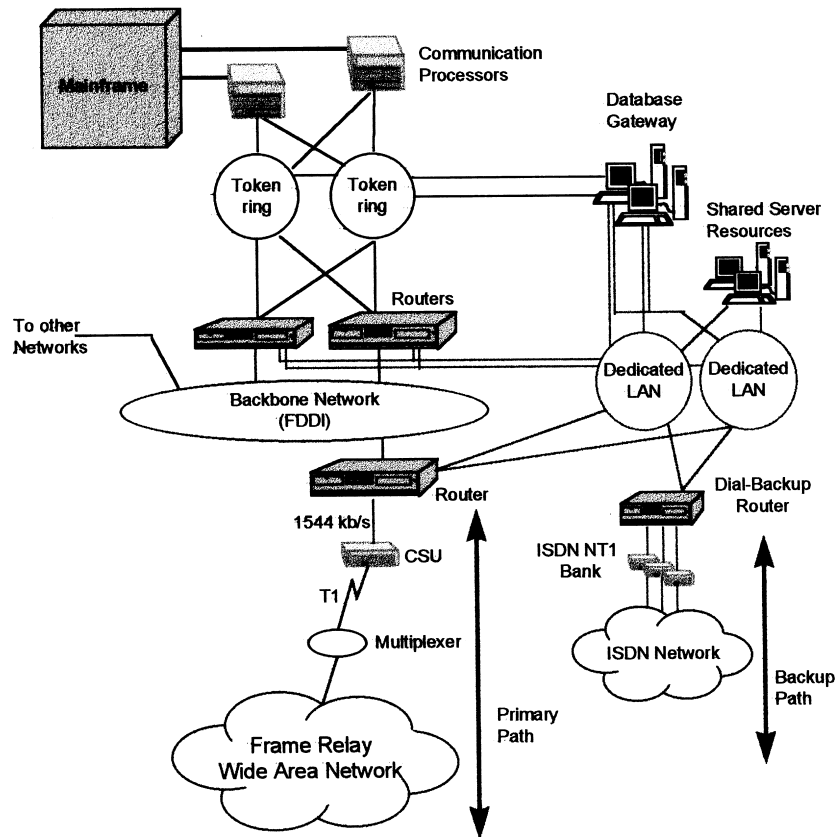


Fig. 17. Central site network design.

the transmission of supervisory frames (such as data acknowledgment and “keep alive” overhead traffic) over the frame relay wide area network.

To minimize the impact of large IP or IPX file transfers on delay sensitive SNA traffic, we need to implement priority mechanisms at the frame relay access interface of the central, hub, and user site routers. Priority mechanisms provide a combination of traffic flow separation, per-flow packet queuing, and packet scheduling disciplines. In the application environment under consideration, we implement priority mechanisms at the frame relay access router interfaces, with the mainframe access related traffic configured for high priority. This approach ensures that SNA sessions will not time-out in the presence of bursty file transfer traffic. If a guaranteed level of service is required for all traffic types, more sophisticated priority mechanisms need to be implemented.

To provide protection against wide area network related failures, we use a *dial backup* mechanism over ISDN (or *switched* 56 kb/s service for the locations where ISDN is not available), which provides a backup connection to the primary frame relay wide area network connection. When a failure occurs in the primary path through the wide area network, the dial-backup connection is established and is used to route traffic to the destination until the primary path recovers. The dial backup mechanism is implemented on routers at the user, hub, and central sites, by properly

configuring a number of interfaces at each router, and placing them in a rotary group. Dialer rotary groups allow dynamic allocation of interfaces to telephone numbers. Before a call is placed or received, the rotary group is searched for an interface that is not in use. The number of physical interfaces that are placed in the rotary group depends on the number of simultaneously active calls that need to be supported. To address security and caller identification issues, we use the Point-to-Point Protocol with Challenge Handshake Authentication Protocol (CHAP) [13]. With CHAP a remote device attempting to connect to a router is requested to respond. When the response is received, the router verifies that the remote device has permission to dial in.

Fig. 17, shows the basic components of the central site network design. As shown in the figure, the network design provides a high degree of redundancy and flexibility, given the importance of the central site to the overall application environment. The proposed design has been implemented and it currently supports around 150 user sites across the US, with plans to expand to 250.

6. Summary

The paper described a business-driven corporate network planning and design methodology that has an iterative

structure to periodically assess, update, and extend the network architecture to support new business applications and requirements, account for changes in business direction, and analyze the implications of technology advances on business processes and functions. The paper focused on the network infrastructure design and analysis task of the proposed methodology and presented the network design and analysis of an application environment characterized by a client/server architecture, integration of legacy and distributed database structures, file and print service support through a common network operating system, support of TCP/IP, SPX/IPX and SNA communication protocols, frame relay wide area network, and token ring local area networks. Specifically, the characteristics of the application workload were analyzed through a series of measurement experiments, and a detailed performance model of the application environment was developed. The model was used to study network performance under various load conditions, identify bottlenecks and compare design alternatives. Finally, a detailed network design was described. The proposed network design has been verified through a series of experiments, and the application environment under consideration has been implemented.

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